

● EASY

● PROBLEM-SOLVING AND DATA ANALYSIS

Ratios, rates, proportional relationships, and units

03

10 questions · ~15 min

Q1

PROBLEM-SOLVING AND DATA ANALYSIS

The ratio x to y is equivalent to the ratio 12 to t . When $x = 156$, what is the value of y in terms of t ?

- A. $13t$
- B. $12t$
- C. $144t$
- D. $168t$

An object travels at a constant speed of 12 centimeters per second. At this speed, what is the time, in seconds, that it would take for the object to travel 108 centimeters?

- A. 9
- B. 96
- C. 120
- D. 972

If $t = 4u$, which of the following is equivalent to $2t$?

- A. $8u$
- B. $2u$
- C. u
- D. $\frac{1}{2}u$

Q4

PROBLEM-SOLVING AND DATA ANALYSIS

A fish swam a distance of 5,104 yards. How far did the fish swim, in miles?

1 mile = 1,760 yards

- A. 0.3
- B. 2.9
- C. 3,344
- D. 6,864

PROBLEM-SOLVING AND DATA ANALYSIS

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

RATIOS, RATES, PROPORTIONAL RELATIONSHIPS, AND UNITS — SET 3 · PROBLEM-SOLVING AND DATA ANALYSIS 6/11

A special camera is used for underwater ocean research. The camera is at a depth of 39 fathoms. What is the camera's depth in feet? 1 fathom = 6 feet

- A. 234
- B. 117
- C. 45
- D. 7

Q7

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

A participant in a bicycle race completes the race with an average speed of 24,816 yards per hour. What is this average speed, in miles per hour? 1 mile = 1,760 yards

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Makayla is planning an event in a 5,400-square-foot room. If there should be at least 8 square feet per person, what is the maximum number of people that could attend this event?

- A. 588
- B. 675
- C. 15,274
- D. 43,200

Q9

PROBLEM-SOLVING AND DATA ANALYSIS

How many yards are equivalent to 612 inches? 1 yard = 36 inches

- A. 0.059
- B. 17
- C. 576
- D. 22,032

A kangaroo has a mass of 28 kilograms. What is the kangaroo's mass, in grams?

(1 kilogram = 1,000 grams)

A. 28,000

B. 1,028

C. 972

D. 784